Chapter test

9 Transport of substances in humans

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

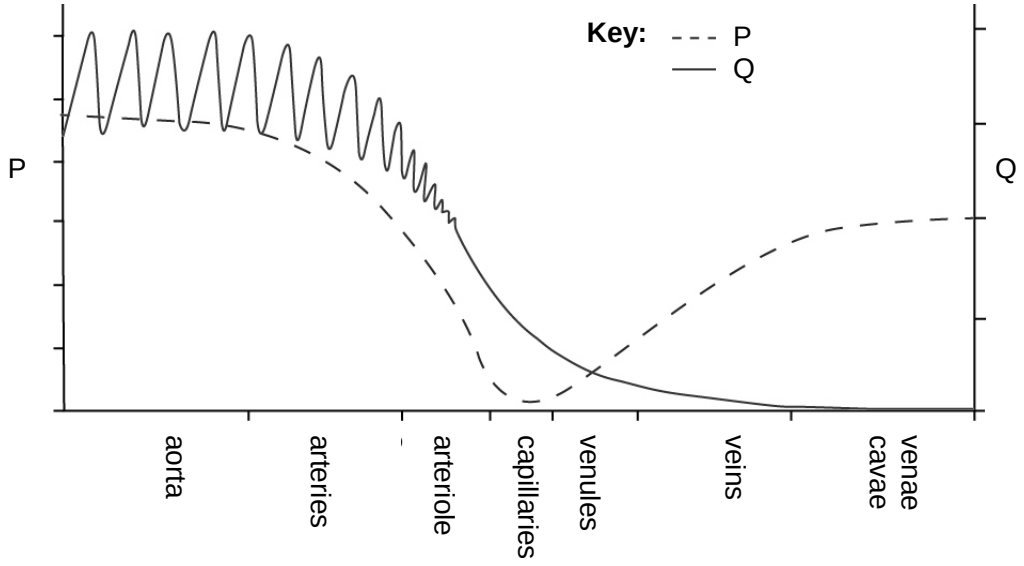
Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following comparisons of red blood cells and phagocytes is ***incorrect***?

	Red blood cells	***Phagocytes***
A.	biconcave disc shape	irregular in shape
B.	fewer in number in the blood	greater in number in the blood
C.	do not have a nucleus	have a nucleus
D.	red in colour	colourless

2. The graph below shows the changes in two parameters, P and Q, along different types of blood vessels in the human circulatory system.



What do P and Q represent respectively?

	P	***Q***
A.	blood pressure	velocity of blood flow
B.	diameter of the vessel	blood pressure
C.	velocity of blood flow	diameter of the vessel
D.	velocity of blood flow	blood pressure

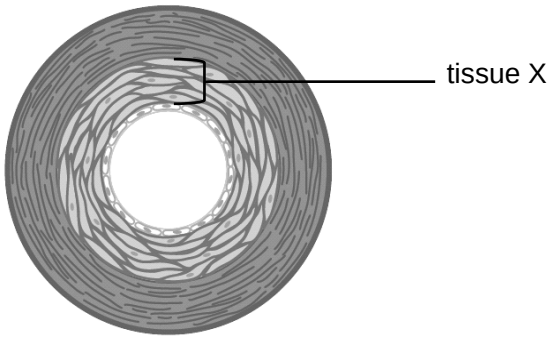
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★ 3. All veins

(1) carry blood towards the heart.
(2) carry deoxygenated blood.
(3) have thin muscle walls.

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only ☐

★ 4. The diagram below shows the cross section of an artery.



What are the functions of tissue X?

(1) It provides force to push the blood along the artery.
(2) It helps the artery withstand high blood pressure produced by the heart.
(3) It regulates the amount of blood flowing through the artery.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

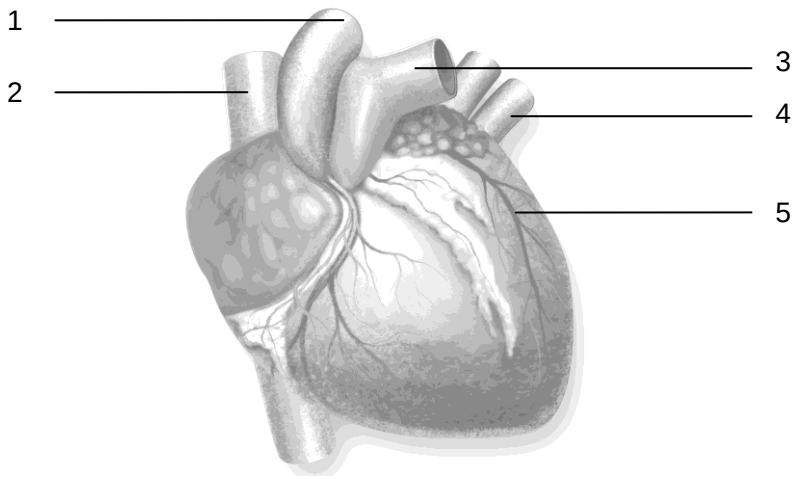
★ 5. Which of the following combinations correctly describes the states of tricuspid valve and semilunar valves when the right ventricle is relaxing?

	Tricuspid valve	***Semilunar valves***
A.	closed	closed
B.	open	closed
C.	open	open
D.	closed	open

☐

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Directions: Questions 6 and 7 refer to the diagram below, which shows the blood vessels connecting the heart.



★ 6. Which of the following blood vessel(s) carry/carries deoxygenated blood?

A. 1 only
B. 2 only
C. 2 and 3 only
D. 3 and 4 only ☐

★ 7. Vessel 5 supplies nutrients to the cardiac muscle. It receives blood from

A. 1.
B. 2.
C. 3.
D. 4. ☐

★ 8. Which of the following characteristics of capillaries allows the formation of tissue fluid?

A. They are in close contact with tissue cells.
B. They are differentially permeable.
C. They have a large cross-sectional area.
D. They have a large surface area. ☐

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★ 9. Which of the following are the functions of the lymphatic system?

(1) To filter bacteria.
(2) To return tissue fluid back into the blood circulation.
(3) To absorb and transport lipids.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★★ 10. P, Q, R and S are four different kinds of fluid in a human body. According to the descriptions, identify P, Q, R and S.

	Presence of red blood cell	**Presence of large protein molecules**	**When compared with another fluid**
P	✗	✗	The pressure of P is higher than S.
Q	✓	✓	The pressure of Q is lower than R.
R	✓	✓	The pressure of R is higher than P.
S	✗	✗	The pressure of S is lower than R.

	P	***Q***	***R***	***S***
A.	lymph	blood entering capillary bed	tissue fluid	blood leaving capillary bed
B.	lymph	blood leaving capillary bed	blood entering capillary bed	tissue fluid
C.	blood leaving capillary bed	tissue fluid	lymph	blood entering capillary bed
D.	tissue fluid	blood leaving capillary bed	blood entering capillary bed	lymph

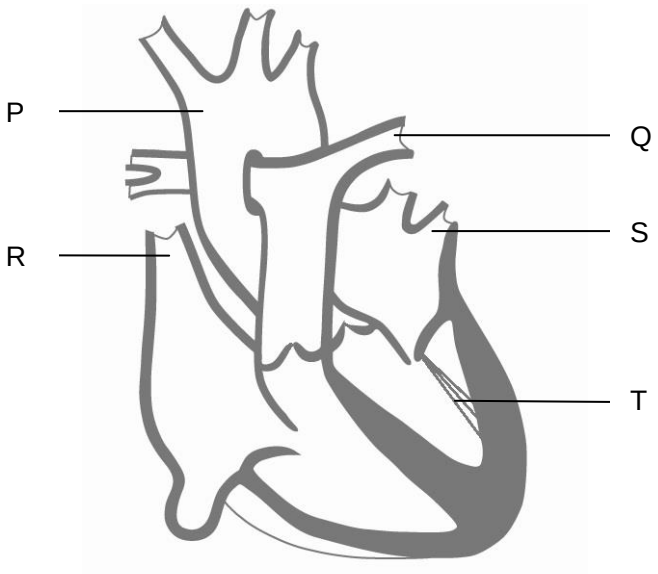
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Question	1	2	3	4	5	6	7	8	9	10
Related section	9.2	9.3	9.3	9.3	9.4	9.4	9.4	9.6	9.7	9.7

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B. Problem solving (20 marks)

★ 11. The diagram shows a section through the human heart.



(a) Using the letters in the diagram, state the blood vessel that carries blood from the heart to the lungs. (1 mark)

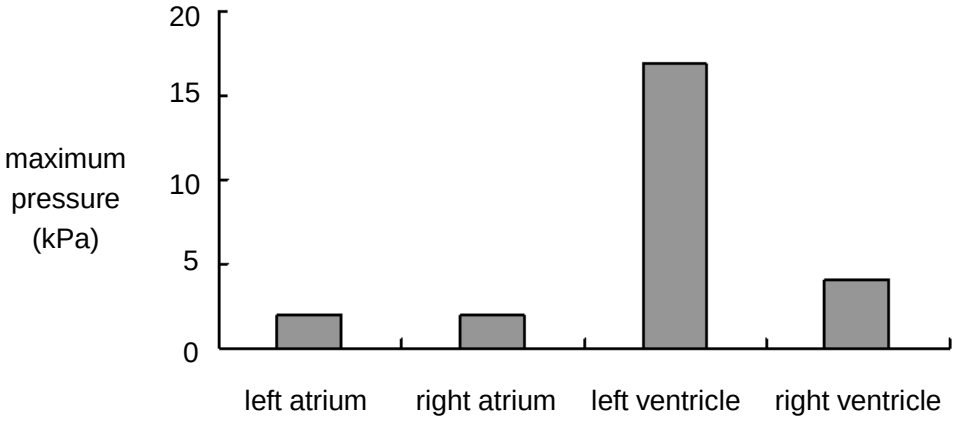
(b) Is the left ventricle relaxing or contracting? Give ***one*** piece of evidence from the diagram to support your answer. (2 marks)

(c) Name structure T. What is the function of structure T within the heart? (2 marks)

(Total: 5 marks)

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★ 12. The bar chart below shows the maximum pressure produced by each heart chamber during contraction.



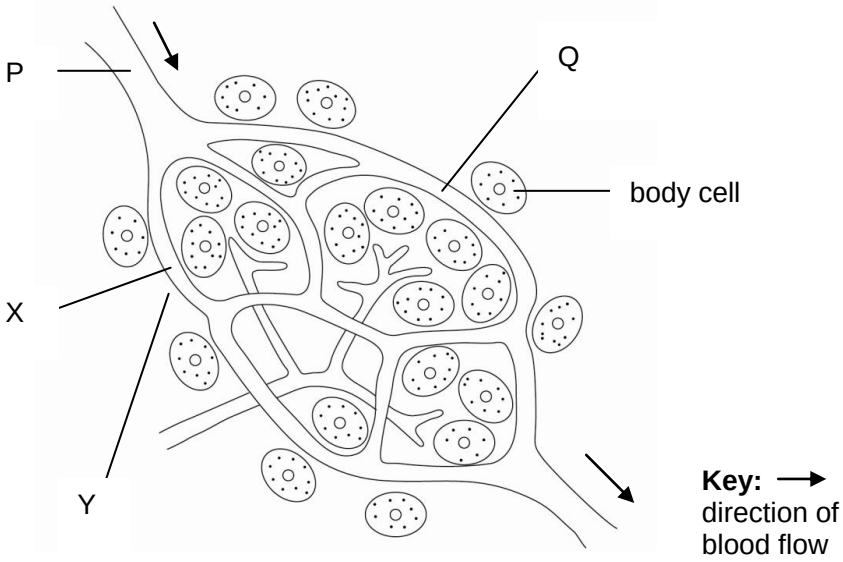
(a) Explain why the maximum pressure of the left and right atria is more or less the same, while there is a great difference in the maximum pressure of the two ventricles. (2 marks)

(b) What is the significance of producing a relatively higher pressure in the left ventricle? (2 marks)

(Total: 4 marks)

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★ 13. The diagram shows a capillary bed and some surrounding cells.



(a) State the vessel type of P. (1 mark)

(b) Give ***one*** way in which the composition of the fluid at X differs from the composition of the fluid at Y. Explain the difference. (2 marks)

(c) How does the structure of vessel Q facilitate the exchange of materials between the blood and body cells? (4 marks)

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(d) Patients with kwashiorkor have a swollen belly due to the accumulation of tissue fluid in the abdomen. Suggest why the accumulation of tissue fluid occurs in persons on a protein-deficient diet. (4 marks)

(Total: 11 marks)

C. Exam challenge (10 marks)

★★14. The table below shows the rate of blood flow in various parts of the body of a student at rest and during vigorous exercise.

Body parts	Rate of blood flow (cm^3/minute)	
	at rest	during vigorous exercise
brain	750	750
skeletal muscles	1200	12500
cardiac muscle	250	750
kidneys	1100	600
skin	500	1800
intestines	1300	600

(a) How does the rate of blood flow to skeletal muscles change during vigorous exercise? What is the significance of such change? (3 marks)

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(b) Under both conditions, the rate of blood flow in the brain remains unchanged. State the significance of this. (1 mark)

(c) As blood passes from the heart to other parts of the body, its composition changes. State and explain ***two*** difference in the composition of blood of renal artery and renal vein, the two main vessels connecting the kidneys. (4 marks)

(d) Explain why it is inadvisable to have vigorous exercise right after a meal. (2 marks)

(Total: 10 marks)

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Marking scheme

A. Concept checking (10 marks)

1.	B	2.	D	3.	C	4.	C	5.	B
6.	C	7.	A	8.	B	9.	D	10.	D

(1 mark each)

B. Problem solving (20 marks)

11. (a) Q (1)

(b) The left ventricle is relaxing. (1)

The semilunar valve in vessel P (aorta) closes / the bicuspid valve opens when the left ventricle relaxes. (1)

(c) heart tendon (1)

It prevents the heart valves from turning inside out (into the atrium) when the ventricle contract.(1)

(Total: 5 marks)

12. (a) The muscle walls of the two atria are of a similar thickness. They contract to produce a force of a similar magnitude to pump blood into the ventricles. (1)

The muscle wall of the left ventricle is much thicker than that of the right ventricle. Thus the contraction force is much higher in the left ventricle. (1)

(b) A stronger force is produced (1)

to pump the blood to different parts of the body. (1)

(Total: 4 marks)

13. (a) arteriole (1)

(b) The fluid in X contains red blood cells / plasma proteins / blood platelets, while the fluid in Y does not. (1)

This is because they are too large to pass through the walls of capillary. (1)

(c) Any ***two*** sets of the following: (1)

- The vessel wall is differentially permeable. (1)

  This allows the formation of tissue fluid, which serves as a medium for the exchange of materials between the blood and body cells. (1)

- The vessel wall is one-cell thick. (1)

  This provides a short diffusion distance for the exchange of materials. (1)

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- The lumen is small so that red blood cells come into close contact with the vessel wall. (1)

  This shortens the distance for oxygen to diffuse from red blood cells to the body tissues. (1)

(d) The blood in the person on a protein-deficient diet contains less plasma proteins, (1)

resulting in a higher water potential of the blood. (1)

Thus, the water potential difference between the blood and tissue fluid decreases. (1)

The amount of water in the tissue fluid reabsorbed into the capillaries by osmosis decreases. (1)

(Total: 11 marks)

C. Exam challenge (10 marks)

14. (a) The rate of blood flow increases from 1200 cm^3/minute at rest to 12500 cm^3/minute during vigorous exercise. (1)

The increased blood flow supplies more oxygen and glucose to the skeletal muscles for carrying out respiration (1)

to release more energy for muscular contraction. (1)

(b) Brain cells need a consistent supply of oxygen and glucose for survival. (1)

(c) Any ***two*** sets of the following: (4)

- The blood in renal vein contains less urea than that in renal artery. (1)

  Urea is removed by the kidneys and is excreted in urine. (1)

- The blood in renal vein contains less oxygen / glucose / nutrients than that in renal artery. (1)

  Cells in the kidneys use these substances for respiration / in metabolic processes. (1)

- The blood in renal vein contains more carbon dioxide than that in renal artery. (1)

  Carbon dioxide is formed when the cells in the kidneys carry out respiration. (1)

(d) The rate of blood flow to the intestines decreases during vigorous exercise. (1)

The intestine may not be able to function normally, affecting the digestion and absorption of food. (1)

(Total: 10 marks)

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